Radar range and velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate a radar's range measurement from its line-of-sight velocity measurement.

SUCCESS CRITERIA - I can...

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "round trip" and "signal" to explain your answer. Sentence starters are optional.

round trip: travel from start to a reflector and back

signal: a disturbance that carries information

Doppler shift: a frequency change associated with relative motion

01 Warm-up: bring back an earlier idea

Two EM waves in vacuum have frequencies f and **3** f. **Compare speeds** and wavelengths.

Word help: vacuum = space with negligible matter

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

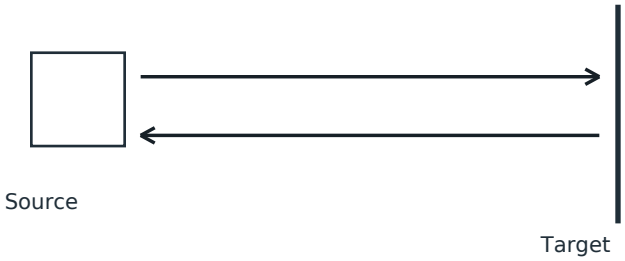
02 Worked example: explain the prediction

Reference: range = c Δt/**2**; c = **3.0×10^{8} m/s**.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

Worked reasoning: A radar echo returns after **2 microseconds**. With c=**3×10^{8} m/s**, range=c Δt/**2**=**300 m**. The measured time includes the outward and return paths.

Your turn: Draw the round trip and **label the conversion 2 microseconds=2×10^{-6} s**.

Sentence starter: In my work, I used … to show … . I labeled it … .

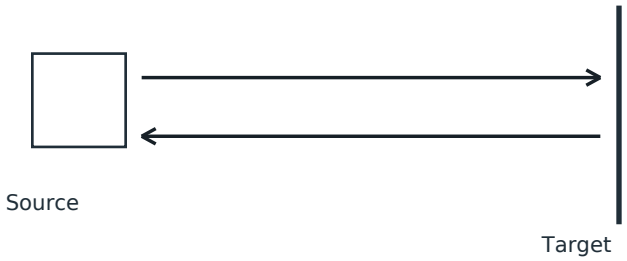


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw outgoing** and returning radar paths.
2. **Which measurement** helps range: echo delay or frequency shift?
3. Which helps radial velocity?

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

04 Apply the physics

A scaled radar animation uses signal speed **20 m/s** and an echo delay of **3 s**. **Label the round trip**. **Find total travel distance** and one-way range. This animation deliberately uses a slower speed than real radar.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary? **Sketch another possible motion**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Add a velocity perpendicular to the line of sight.



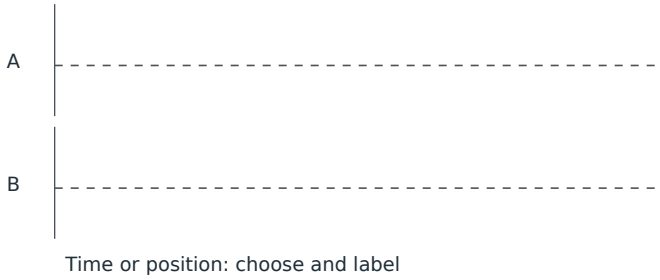
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A wave has frequency **5 Hz** and wavelength **7 m**. **Find its speed**.
Label a wave sketch to **support the calculation or comparison**.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

07 Exit check

An echo’s out-and-back distance represents **1000 m**. **How far** is the target?

Sentence starter: I notice ... . This supports ... because ... .

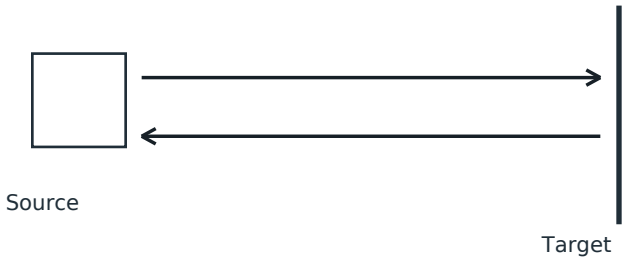


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Echo times change from **4** to **3 microseconds**. **Calculate both ranges** and state whether the target is approaching or receding.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③